

WORKER SAFETY MONITOR

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THE PROBLEM

Construction
sites are
dynamic and 
high risk

44

**Fall from height
fatal & major
injuries**

in singapore construction, 2024

Problem:

**“Falls can happen before anyone
know”**

2

Slip trip & falls
in singapore construction, 2024

PROBLEM STATEMENT :

“How can we automatically **detect fall-from-height** , Slip-Trip-Fall and near-miss events, and **alert authorised personnel** for prompt intervention to improve construction-site safety”

02

PROJECT OBJECTIVE

01

MONITOR

Monitor worker movement in real time using the NESSO N1

02

DETECT

Detect abnormal movement and classify the worker as SAFE, NEAR-MISS, or FALL using rule-based sensor analysis.

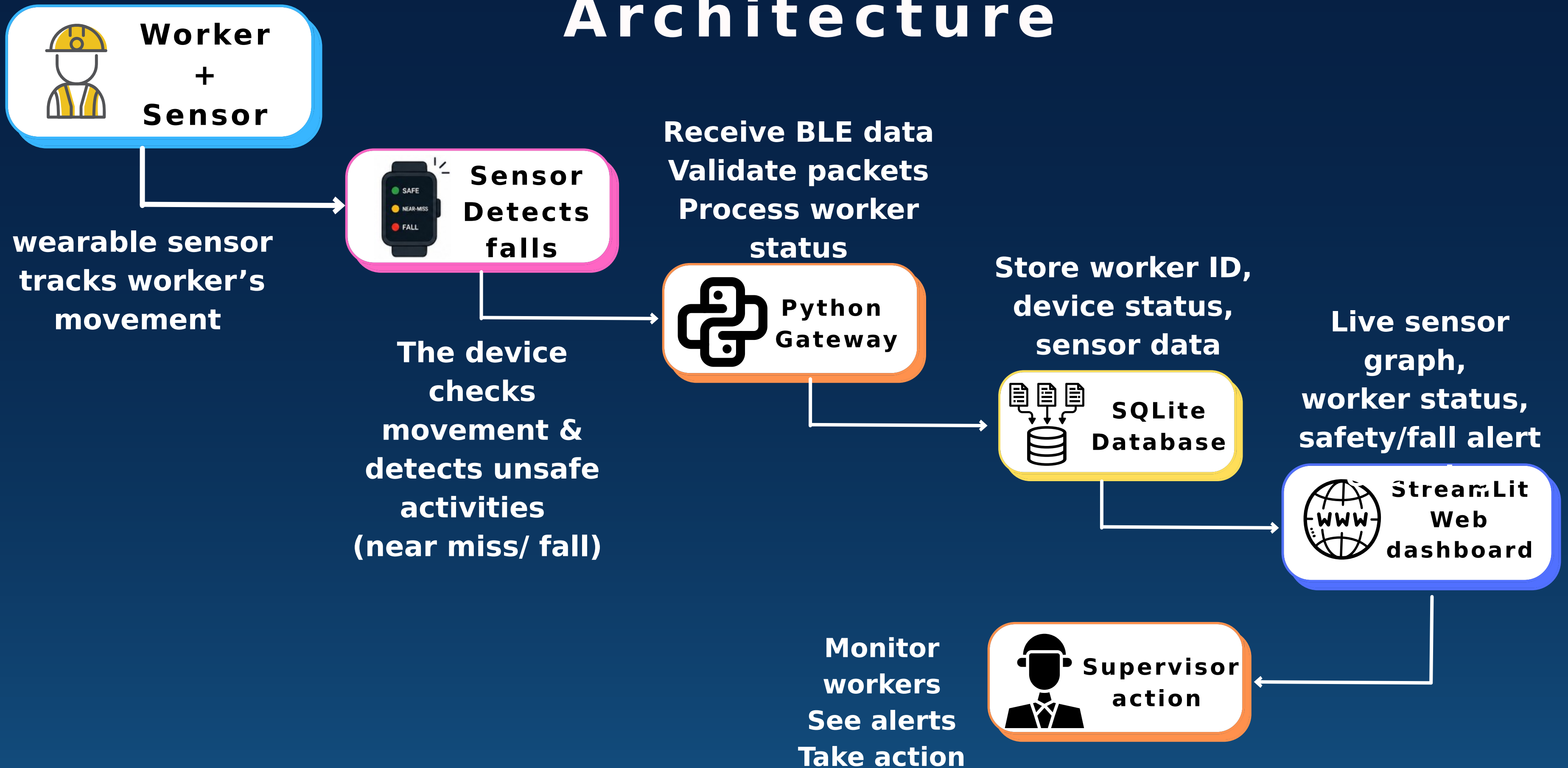
03

ALERT

Provide supervisors with live worker status, sensor graphs, and safety-event alerts through a web dashboard.



Live Web Dashboard Architecture



PROJECT TIMELINE & MILESTONES

	Week 11	Week 12	Week 13	Week 14	Week 15	Week 16	Week 17	Week 18
Data collection	Data Collection							
Data cleaning		Data Cleaning						
Feature extraction			Feature Extraction					
Event detection				Event detection				
Database Integration					Database Integration			
Dashboard Developement						Live web dashboard development		
Review & release							Review & Release	

DATA PIPELINE OVERVIEW

Collect data



- Nesso N1 sensor
- Data from 4 workers

Clean data



- Remove invalid readings
- Check sampling intervals

Analyse data



- Identify key features
- Analyse movement trend

Setting threshold detection



- Determine thresholds for events
- Classify

Validate



- Test against labelled data
- Evaluate detection threshold accuracy

Database storage



- Store sensor data & detected events

Visualisation



- Real time dashboard
- alerts for falls & near misses
- event log

ASSUMPTIONS

01 — SENSOR PLACEMENT

Assume the NESSO is worn in a consistent position and orientation, so movement readings are comparable between tests.

02 — TEST DATA REPRESENTATION

Assume the controlled normal, near-miss and fall-like movements from the 4 workers are representative enough to develop and test the prototype detector.

03 — SAMPLING RATE

Assume 25 Hz is sufficient to capture the important acceleration and gyroscope changes needed for safety-event detection.



DATA COLLECTION

DATA SIZE DECISION

WHY 25 HZ?

Sampling Rate	Time per Reading	Data Volume	BLE / Processing Load	Suitability for Prototype
100 Hz	10 ms	Very High	High	More data than needed; heavier processing
50 Hz	20 ms	Medium	Medium	Good, but still more data than required
25 Hz	40 ms	Low	★ Lower	Good balance for live monitoring

- Sufficient resolution enough to capture sudden movement
- limit unnecessary data generation while still provide enough readings for event detection
- reduces BLE traffic and processing requirements, making the system more manageable when monitoring four workers

=

25 Hz

Most appropriate

★

DATA COLLECTION PROCEDURE

1) Sensor setup

(NessoN1 Motion Sensor)



- 25 Hz sampling rate configuration

2) Collection Protocol

Protocol



Sensor attached to 4 workers



Each Worker performs

- Normal movement
- Near-Miss movement
- Fall-Like movement

1.Raw Data sample collection

timestamp	elapsed_seconds	interval_ms	Xg	Yg	Zg	Xdeg
2026-08-03T12:25:50.573	0.042097		0.1500	-0.1000	0.9300	29.9700
2026-08-03T12:25:50.602	0.071057	28.959500	0.3400	-0.1500	1.0000	8.9100
2026-08-03T12:25:50.646	0.115695	44.638400	0.4200	-0.1900	1.0500	11.6600
2026-08-03T12:25:50.691	0.160931	45.235500	0.3800	-0.1800	1.0500	13.6100
2026-08-03T12:25:50.721	0.190885	29.954200	0.3700	-0.1700	1.0300	4.7000
2026-08-03T12:25:50.769	0.238483	47.598000	0.4000	-0.2000	1.0200	-0.7300
2026-08-03T12:25:50.811	0.280872	42.389200	0.3700	-0.1900	1.0100	-1.7700

- 12 x 3 raw (csv) recordings were automatically **collected** and **stored** using python



DATA CLEANING

DATA CLEANING

1

standardized columns



Convert different column names into one format

Xg, Yg, Zg, Xdeg, Ydeg, Zdeg

2

Validate Structure



- Check all 6 required sensor columns exist
- Reject unusable files

3

Handle Missing / Invalid Data



- Convert sensor values to numeric
- Remove rows with missing required values

4

Remove Bad Records



- Remove duplicate rows
- Remove corrupted sensor readings

5

SAVE & ORGANISE DATA



- Sort by time
- Add worker/activity labels
- Save cleaned files
- Combine into one cleaned dataset

source_file	worker	recording_type	rows_before	rows_after	rows_removed	missing_rows_removed	duplicate_rows_removed
(die2) HJ motion_data.csv	hong_jean	fall_like	3000	3000	0	0	0
(near miss) HJ motion_data.csv	hong_jean	near_miss	3000	3000	0	0	0
motion_data (hj).csv	hong_jean	normal_work	2993	2993	0	0	0
KT_Fall_cleaned.csv	kwanteng	fall_like	1703	1703	0	0	0
KT_NearMissed_cleaned.csv	kwanteng	near_miss	1567	1567	0	0	0
KT_Normal.csv	kwanteng	normal_work	5251	5251	0	0	0
(die) pieere.csv	pierre	fall_like	3001	3001	0	0	0
(near miss) P.csv	pierre	near_miss	3000	3000	0	0	0
(normal) P motion_data.csv	pierre	normal_work	3005	3005	0	0	0
ziquan_die_data.csv	ziquan	fall_like	2897	2897	0	0	0
ziquan_near_miss_die.csv	ziquan	near_miss	2896	2896	0	0	0
ziquan_normal_work_data.csv	ziquan	normal_work	2908	2908	0	0	0

DATA ANALYSIS

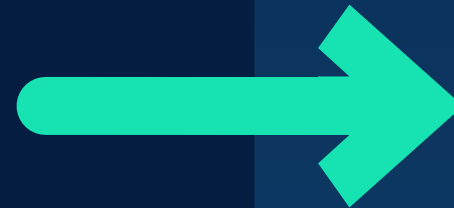


DATA ANALYSIS OVERVIEW

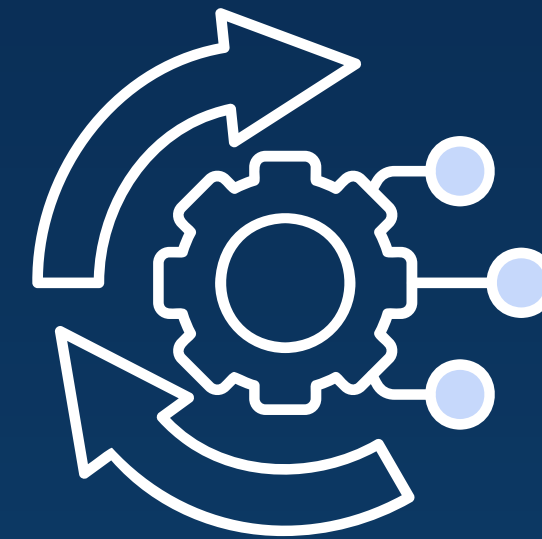
Analyse movement



- Split data into time windows
- Compare Normal movement vs falls/near miss data



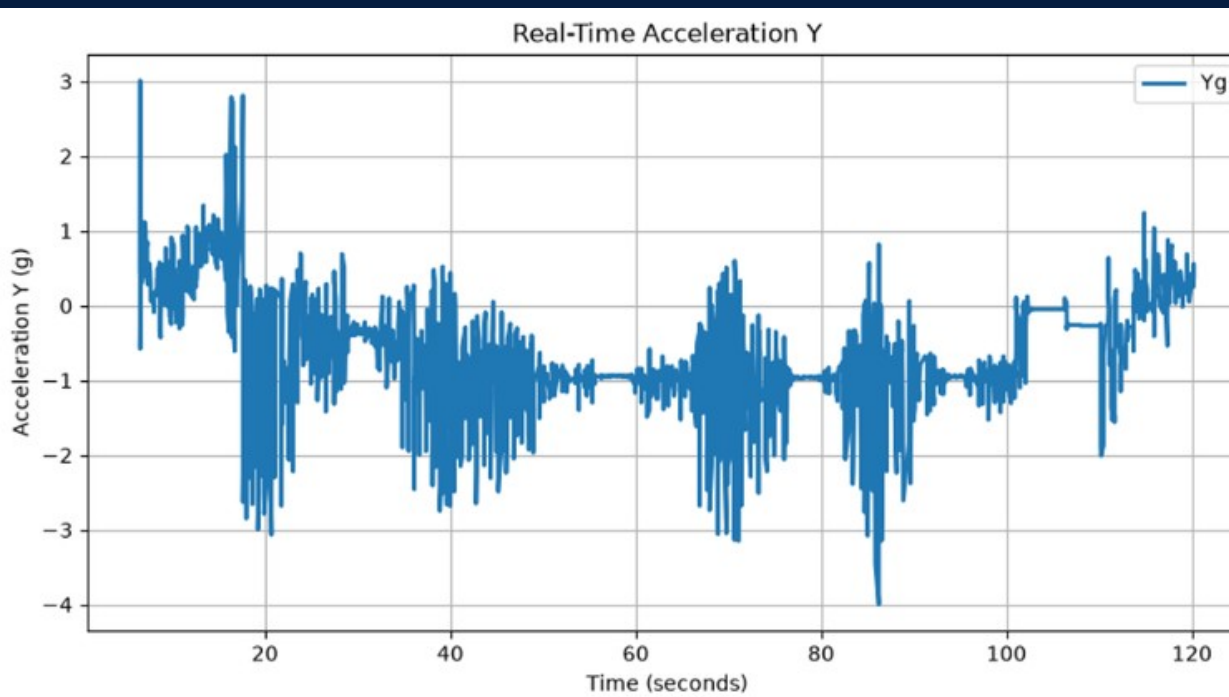
Find key features



- Extract important movement features
- identify useful signals

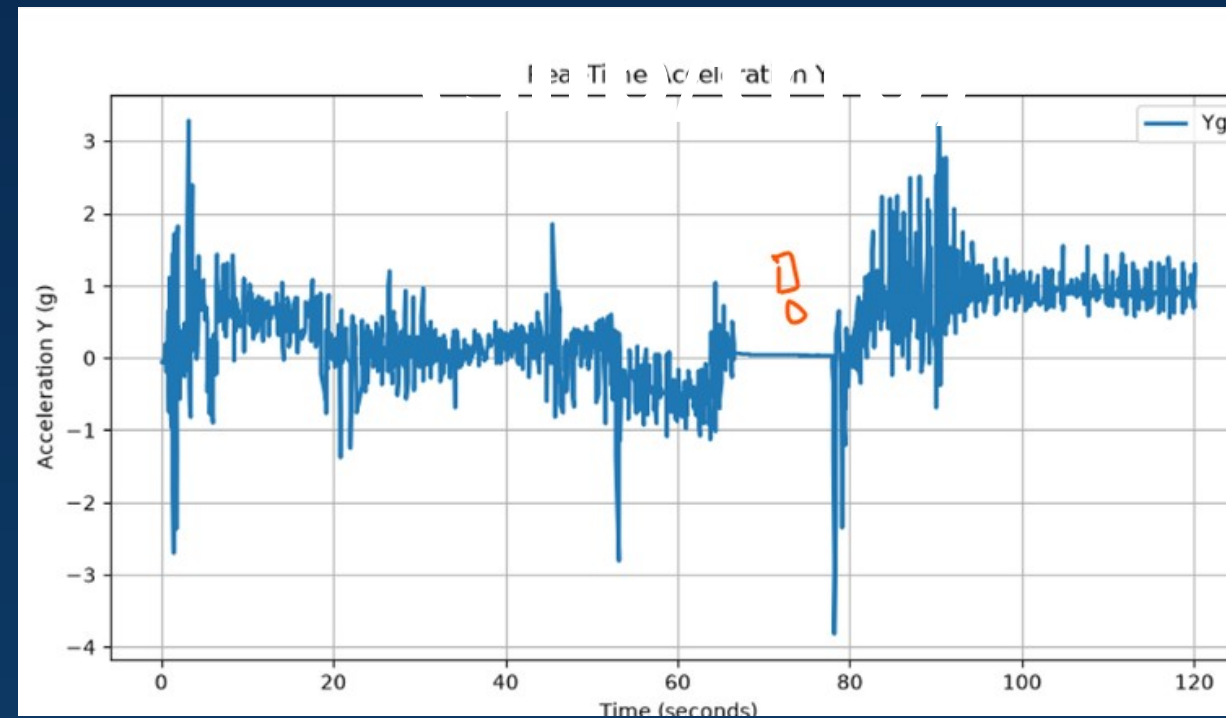
ANALYSE MOVEMENT PATTERN

Normal



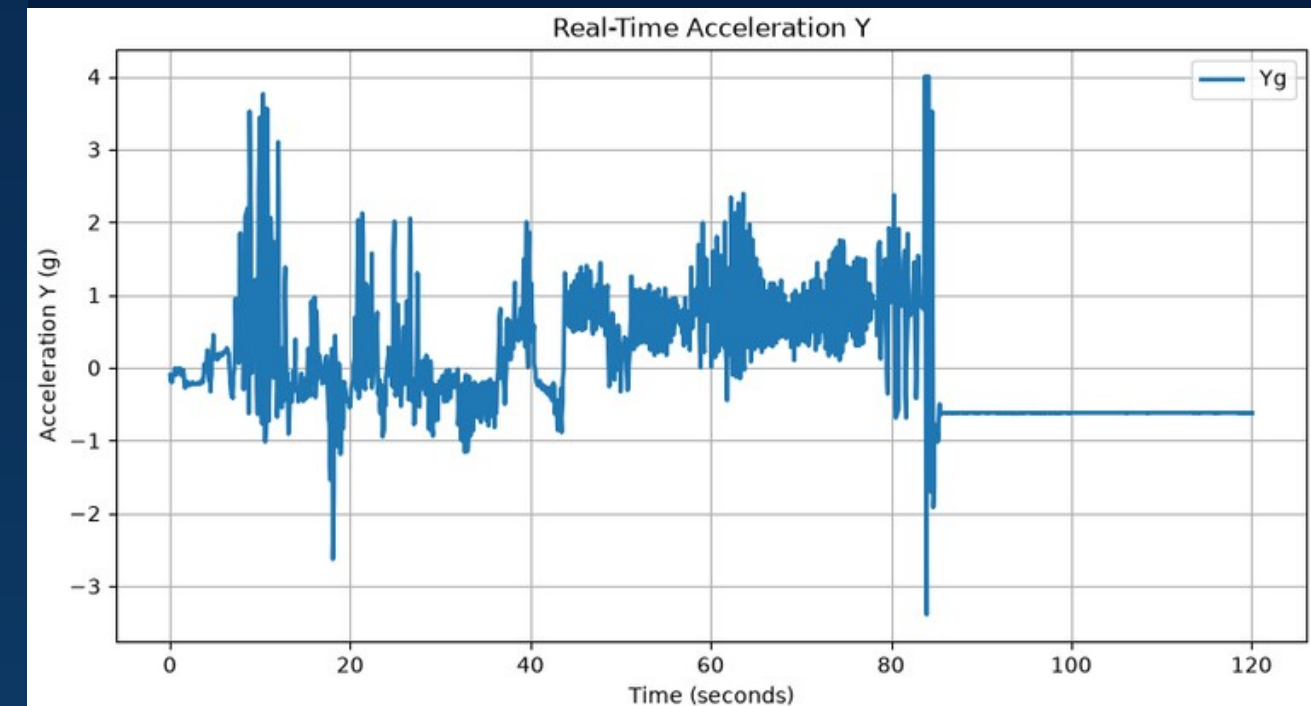
**Smooth / regular
movement**

Near miss



**Sudden movement
change**

Fall



**Large / abrupt
movement**

FIND KEY FEATURES (REDUCE)

6 raw sensor axes

Accelerometer

Gyroscope

Xg	Yg	Zg	Xdeg	Ydeg	Zdeg
Acceleration along X direction	acceleration Y direction	acceleration Z direction	Rotation around the X direction	Rotation Y direction	Rotation Z direction

→ Process & combine = from 6 to 4 key movement signals

Acceleration Magnitude

Overall movement / impact force

$$A = \sqrt{x^2 + y^2 + z^2}$$

Gyroscope Magnitude

Overall rotation speed

$$G = \sqrt{x^2 + y^2 + z^2}$$

Jerk

How sudden the movement change

$$J = |\Delta A| / \Delta t$$

Rotation Activity

Accumulated rotation during event

$$R = \int G \, dt$$

SETTING THRESHOL D

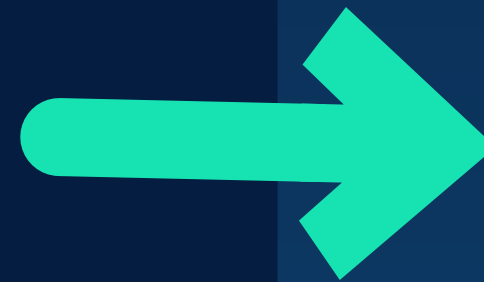


SETTING THRESHOLD OVERVIEW

Test threshold



- **Identify and set threshold values**



Create detection rule



- **Combine 4 key signals to create safety event**

TEST & IDENTIFY THRESHOLD

Collect data

Collect data of:

- **normal**
- **near-miss**
- **fall**

**Calculate the 4
key signals**

**For every recorded
event:, we calculate
and observe:**

- acceleration
magnitude
- Gyroscope
magnitude
- jerk
- rotation activity

For example:

- **Fall** = high
acceleration...
- **Normal** = low
accleration

**Compare
values**

**Choose values that
detect
unusual/dangerous
movement while
avoiding normal
movement.**

**Set the
threshold**

FINAL THRESHOLD CONFIGURATION

$\geq 3.28 \text{ g}$

**ACCELERATION
MAGNITUDE**

Overall movement
/ impact
force

$\geq 641.29 \text{ dps}$

**GYROSCOPE
MAGNITUDE**

Overall rotation
speed

≥ 67.41
g/s

JERK

Sudden
acceleration
change

$\geq 524.51^\circ$

**ROTATION
ACTIVITY**

Accumulated
rotation

L	M	N	O
threshold_event_peak_acceleration_g	threshold_event_peak_gyroscope_dps	threshold_event_peak_jerk_g_per_s	threshold_event_rotation_integral_deg
3.281536	641.286497	67.407807	524.512582

DETECTION RULE/SYSTEM

- Combine 4 key signals to create safety event

trigger :

ACCELERATIO
N
MAGNITUDE

GYROSCOP
E

MAGNITUDE

JER
K

ROTATIO
N
ACTIVITY

- Detection = rule

- If 2/3 out of the 4 signals exceed their threshold : Near miss event trigger
- If 4 out of the 4 signals exceed their threshold : fall event trigger

VALIDATION OF THRESHOL D



VALIDATION OF THE THRESHOLD

01 — Freeze the thresholds

No further threshold changes during validation.

02 — Replay full recordings

The detector scanned the complete recording, not only the known event moment.

03 — Compare with labelled incidents

Alerts were checked against the confirmed event locations.

04 — Record every result

Detection results and normal alerts were saved into a validation CSV.

Main results

7 / 7

RELIABLE INCIDENTS DETECTED

All reliably labelled abnormal events were detected.

0

FALSE NEGATIVES

No reliable labelled incident was missed.

1 / 4

NORMAL RECORDINGS PRODUCED AN ALERT

One normal recording triggered the detector.

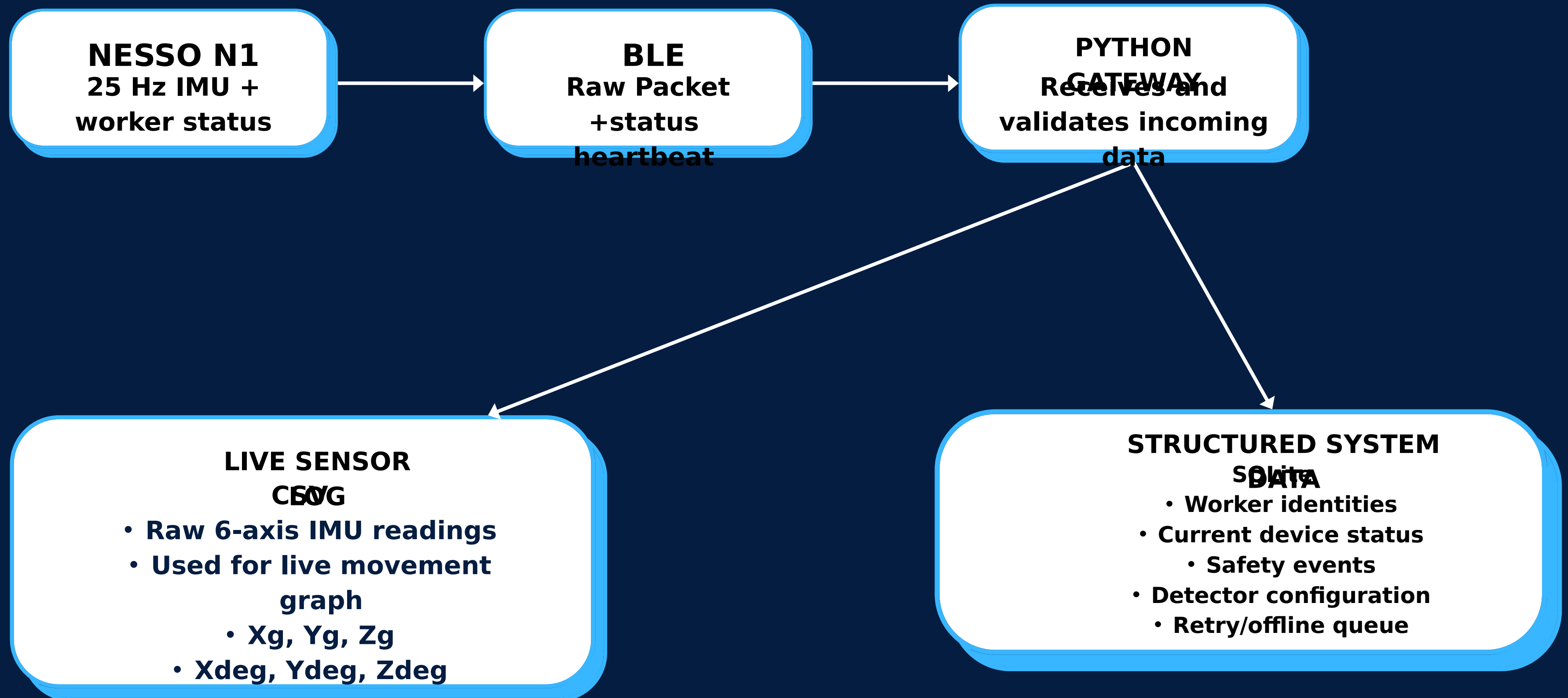
worker	source_file	recording_type	suggested_candidate_rank	suggested_event_time_seconds	verification_score	score_gap_to_second	confidence	posture_c	event_peak	event_peak_g	verification_logic	human_confirmation
hong_jean	(die2) HJ motion_data.csv	fall_like	1	84.16	1	0.6921	HIGH	158.72	6.328	1110.8	fall_like: impact + rotation + jer	YES
hong_jean	(near miss) HJ motion_data.csv	near_miss	2	1.57	0.7973	0.4825	HIGH	133.05	4.577	1144.88	near_miss: disturbance + rotati	YES
kwanteng	KT_Fall_cleaned.csv	fall_like	1	16.17	0.6	0.1924	MEDIUM	13.99	3.045	1129.87	fall_like: impact + rotation + jer	YES
kwanteng	KT_NearMissed_cleaned.csv	near_miss	1	16.53	0.6072	0.3013	HIGH	9.89	4.015	1130.87	near_miss: disturbance + rotati	YES
pierre	(die) pieere.csv	fall_like	2	8.83	0.6704	0.5011	HIGH	83.31	4.04	800.14	fall_like: impact + rotation + jer	YES
pierre	(near miss) P.csv	near_miss	1	9.12	1	0.6623	HIGH	25.84	4.042	547.42	near_miss: disturbance + rotati	YES
ziquan	ziquan_die_data.csv	fall_like	2	33.57	0.6799	0.0008	LOW	4.81	2.42	387.54	fall_like: impact + rotation + jer	YES
ziquan	ziquan_near_miss_die.csv	near_miss	2	58.46	0.7811	0.0584	LOW	68.52	4.881	969.9	near_miss: disturbance + rotati	YES

DATABASE STORAGE



DATA STORAGE ARCHITECTURE

FROM LIVE SENSOR DATA TO STRUCTURED STORAGE



SQLITE DATABASE DESIGN

workers

Worker & Device
Identity

- worker_id
- worker_name
- device_id
- BLE identity
- active status

device_status

Current Live State

- connection status
- safety status
- battery %
- last seen time
- last event time

EVENTS

Safety Event
History

- event_uuid
- worker ID
- device ID
- event time
- event type
- votes
- acceleration peak
- gyroscope peak
- jerk peak
- rotation

Detector_config

- Configuration threshold
- acceleration threshold
- gyroscope threshold
- jerk threshold
- rotation threshold
- required votes
- startup grace
- cooldown

OFFLINE_QUEUE

Gateway Retry
Support

- queued payload
- retry information
- local-first recovery

STORAGE RELIABILITY & RETRIEVAL

HOW WE KEEP SAFETY RECORDS RELIABLE

1. VALIDATE IDENTITY

- worker_id + device_id
- are checked against the workers table.

2.

STORE

Save valid data into SQLite

3. PREVENT DUPLICATES

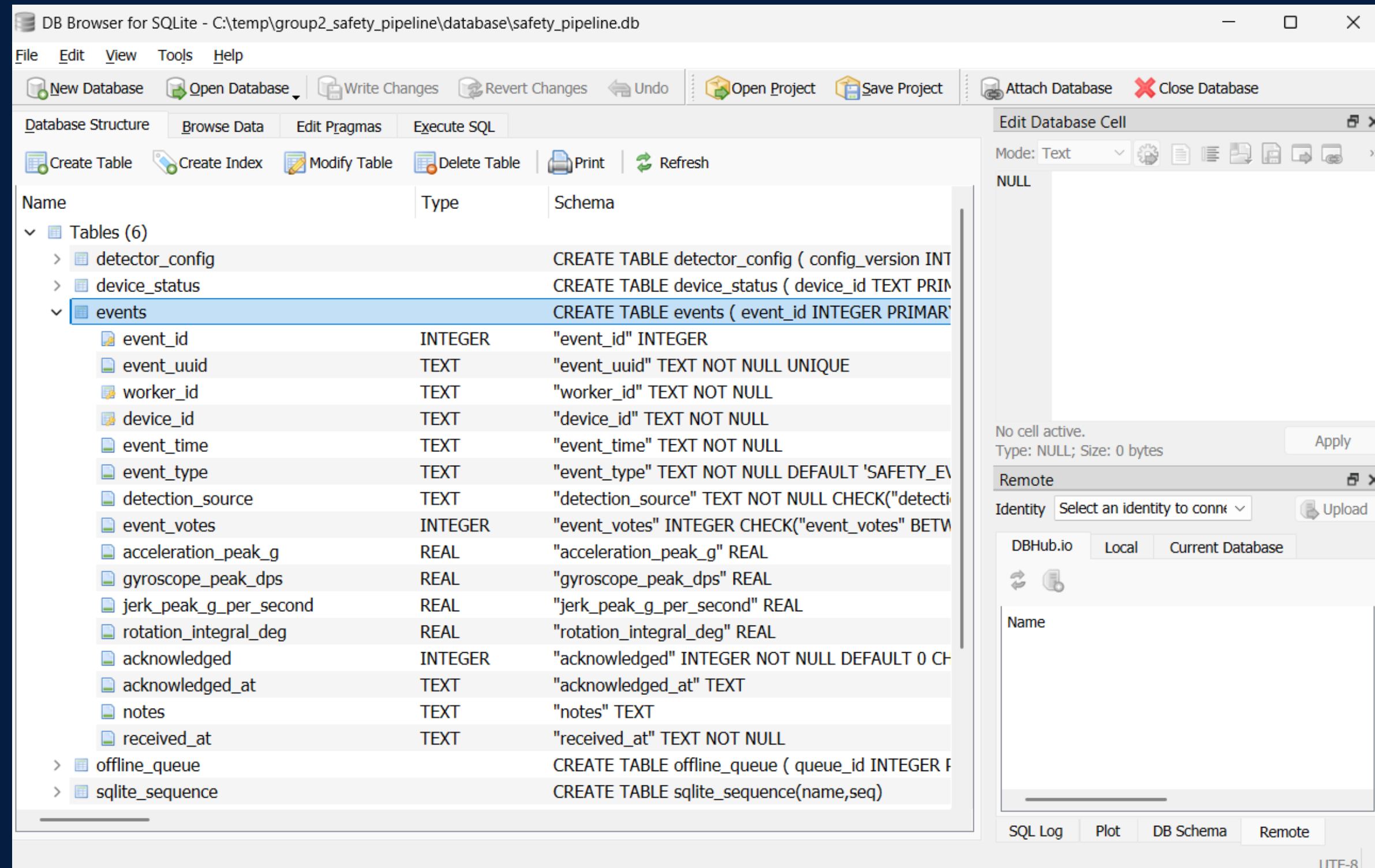
Same event received again → ignored

4.

RETRIEVE

The Streamlit dashboard reads SQLite to show:

- Current worker status
 - Event history
- Detector configuration



DATA VISUALISATION N



ARDUINO NESSO



WEBSITE

☐ Show test / simulated data [?](#)

NESSO Safety Monitor

Real-time worker safety monitoring · EG2A17 Group 5 · Local-first prototype

No fresh NESSO status is being received right now. Start the BLE gateway to begin live monitoring.

[Overview](#) [Incident Log](#) [Devices](#) [Detector](#)

Workers

4

Connected

0

Safety Events

3

Pending Queue

0

✓ All connected workers are currently below the incident threshold.

Workers

Live state refreshes every second from the same four signal thresholds used below. 0–1 votes = SAFE, 2–3 = NEAR-MISS, 4 = FALL.

Hong Jean

HJ · HJ_N1

• DISCONNECTED

BATTERY

—

QUEUE

0

LAST SEEN

No signal yet

LAST EVENT

Never

Kwanteng

KT · KT_N1

• DISCONNECTED

BATTERY

—

QUEUE

0

LAST SEEN

No signal yet

LAST EVENT

Never

Pierre

PI · PI_N1

• DISCONNECTED

BATTERY

—

QUEUE

0

LAST SEEN

No signal yet

LAST EVENT

Never

Ziqian

ZQ · ZQ_N1

• DISCONNECTED

BATTERY

—

QUEUE

0

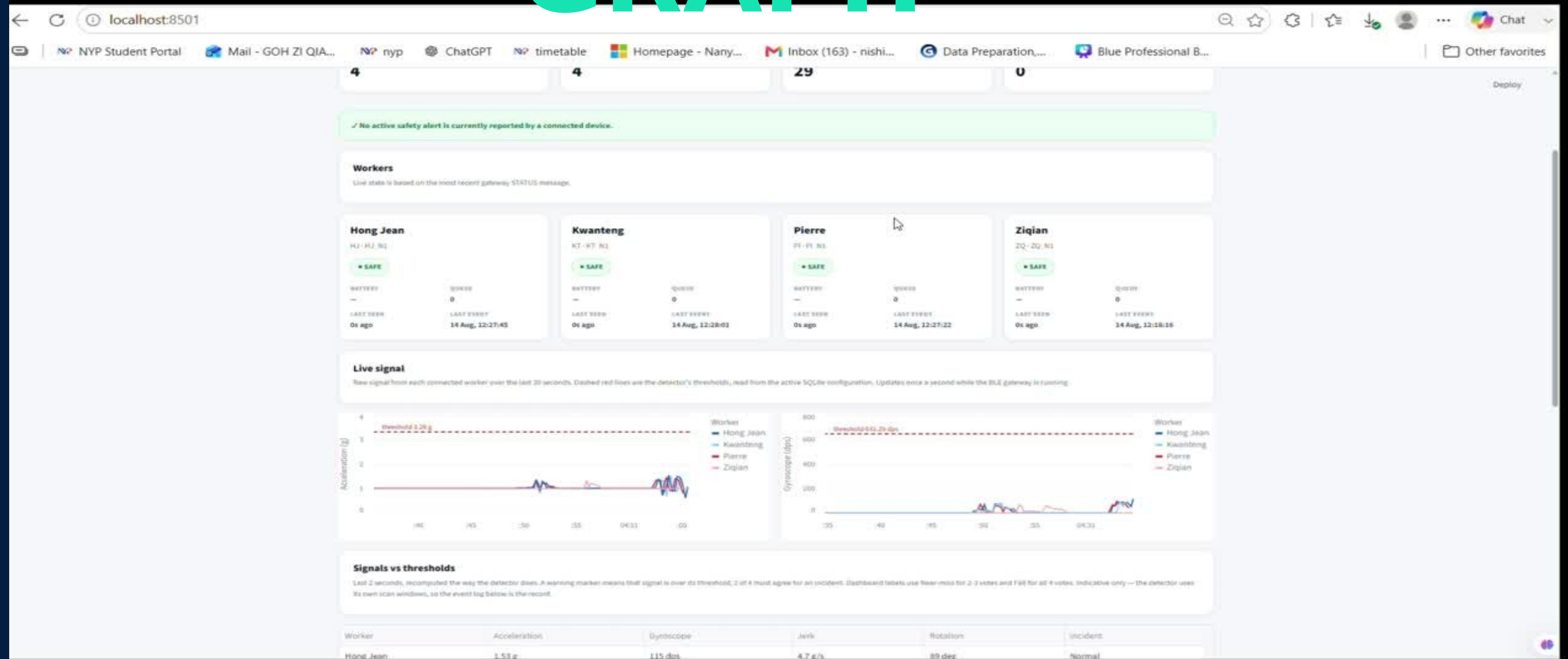
LAST SEEN

37h ago

LAST EVENT

14 Aug, 11:09:07

REAL TIME GRAPH



DATA

☐ Show test / simulated data ?

NESSO Safety Monitor

Real-time worker safety monitoring - EG2A17 Group 5 - Local-first prototype

Overview **Incident Log** Devices Detector

Incident Log

Time	Worker	Device	Event	Peak accel (g)	Peak gyro (dps)	Peak jerk (g/s)	Rotation activity (deg)	Source	Acknowledged	Notes
14 Aug, 12:06:21	Hong Jean	HJ_N1	Near-miss	4.7054	542.1773	76.7843	261.3898	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 12:06:13	Hong Jean	HJ_N1	Near-miss	5.4738	700.6959	67.7029	276.8601	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 12:06:12	Ziqian	ZQ_N1	Near-miss	5.7098	656.8273	61.2855	227.2609	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 12:06:05	Hong Jean	HJ_N1	Fall	5.9471	1131.1422	68.1272	657.5255	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 12:05:56	Ziqian	ZQ_N1	Near-miss	5.1675	1047.5858	58.7814	368.7411	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 12:04:59	Hong Jean	HJ_N1	Near-miss	3.4296	732.2986	44.192	353.5296	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 12:04:44	Ziqian	ZQ_N1	Near-miss	2.917	982.621	34.3144	562.7662	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 12:04:20	Hong Jean	HJ_N1	Near-miss	3.803	684.0122	48.5115	242.6234	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 12:04:20	Ziqian	ZQ_N1	Near-miss	3.8546	670.6932	66.2356	204.6756	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 11:50:40	Ziqian	ZQ_N1	Near-miss	5.4187	1122.5146	41.621	374.6512	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va
14 Aug, 11:50:29	Ziqian	ZQ_N1	Near-miss	4.8283	1031.4259	54.2334	297.9687	BLE_GATEWAY	No	Rule-based 25 Hz detector in Python BLE gateway; va

Rotation activity is integrated gyroscope magnitude; it is not net body angle.

DEMO OF THE ARDUINO NESSO/WEBSITE



FUTURE IMPROVEMENT

Deploy

NESSO Safety Monitor

Real-time worker safety monitoring - EG2A1T Group 5 - Local-first prototype

Overview

Incident Log

Devices

Detector

✓ All connected workers are currently below the incident threshold.

Workers

Live state refreshes every second from the same four signal thresholds used below: 0-1 votes = SAFE, 2-3 = NEAR-MISS, 4 = FALL.

Kwanteng

KT - KT_N1

• SAFE

BATTERY

—

QUEUE

0

LAST SEEN

1s ago

LAST EVENT

Never

Pierre

PI - PI_N1

• SAFE

BATTERY

—

QUEUE

0

LAST SEEN

1s ago

LAST EVENT

Never

Live signal

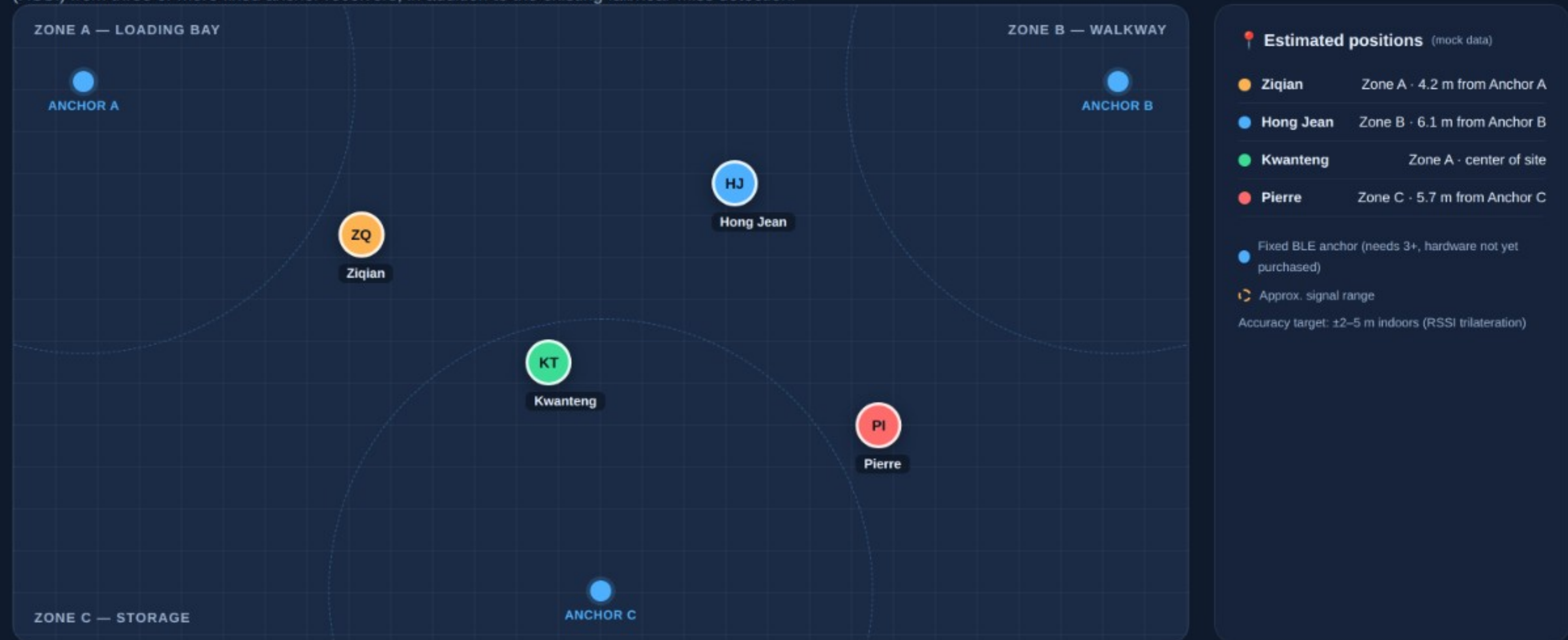
Recent sensor magnitude. Dashed lines are the active detector thresholds.

FUTURE IMPROVEMENT

FUTURE WORK · NOT YET BUILT

Indoor Location Tracking — Concept

Proposed extension to the NESSO Safety Monitor: estimate each worker's position on-site using BLE signal strength (RSSI) from three or more fixed anchor receivers, in addition to the existing fall/near-miss detection.



⚠️ **Concept mockup only** — all positions above are simulated. Real version requires 3+ physical BLE anchors at known fixed points; current system has one gateway.

CONCLUSION



Thank

You

